

# Homework 5 - Theory

Date

Time

1a  $\Sigma_{3,4} = 0 \Rightarrow X_3$  and  $X_4$  are not correlated

1b  $Q_{3,4} = 0 \Rightarrow X_3$  and  $X_4$  are not conditionally correlated

1c The Markov blanket of  $X_2$  are all non-zero values of row 2 that are not self correlated - ex:  $Q_{22}$  is self correlated and does not count,  $Q_{23}$  and  $Q_{24}$  are 0 and therefore not correlated. Therefore

$$\text{Markov blanket} = Q_{21} = X_1$$

1d  $Y_1 = X_1 + X_4 = 1 \ 0 \ 0 \ 1$

$$Y_2 = X_2 - X_4 = 0 \ 1 \ 0 \ -1$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$A\Sigma = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0.71 & -0.43 & 0.43 & 0 \\ -0.43 & 0.46 & -0.26 & 0 \\ 0.43 & -0.26 & 0.46 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.71 & -0.43 & 0.43 & 0.2 \\ -0.43 & 0.46 & -0.26 & -0.2 \end{bmatrix}$$

$$A\Sigma A^T = \begin{bmatrix} 0.71 & -0.43 & 0.43 & 0.2 \\ -0.43 & 0.46 & -0.26 & -0.2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$$

Covariance matrix of  $Y = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$

# HW5 - Theory p2

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$$x^{(1)} = -1, x^{(2)} = 1, \mu_1 = -2, \mu_2 = 2$$

2a E-Step:  $x^{(1)}$  distance to  $\mu_1 = (-1+2)^2 = 1$   
 $x^{(1)}$  distance to  $\mu_2 = (-1-2)^2 = 9$

$$w_1^{(1)} = \frac{e^{-0.5}}{e^{-0.5} + e^{-4.5}} = \frac{1}{1+e^{-4}} \approx 0.98$$

$$w_2^{(1)} = 0.02$$

Symmetry for  $x^{(2)}$ :

$$w_1^{(2)} = 0.02$$

$$w_2^{(2)} = 0.98$$

M-Step:  $\mu_1^{\text{new}} = \frac{w_1^{(1)} x^{(1)} + w_1^{(2)} x^{(2)}}{w_1^{(1)} + w_1^{(2)}} = \frac{0.98(-1) + 0.02(1)}{1.0} = -0.96$

Symmetry for  $\mu_2^{\text{new}} = 0.96$

2b No because Gaussian components have a fixed variance of 1. Each data point is always slightly effected by the other so it never fully converges to -1 and 1.

2c For  $\mu_1 = \mu_2 = 2$  both centers at the same spot give  $w = 0.5$

The weighted average  $\frac{-1+1}{2} = 0$  means both get stuck at 0

2d  $x^{(1)} = -1$  and is closer to  $\mu_1 = -2$  so is assigned to cluster 1

$x^{(2)} = 1$  and is closer to  $\mu_2 = 2$  so is assigned to cluster 2

Update means:  $\text{mean}\{-1\} = -1, \text{mean}\{1\} = 1$

fixed point:  $\mu_1 = -1, \mu_2 = 1$